SI 140A.02 Probability & Statistics for EECS, Fall 2026
Homework 7

Name: Zhang Yiheng
Student ID: 2025531068

Due on Nov. 8, 2026, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to submit your homework in PDF format generated by **LaTeX**, otherwise you will get no points of this homework.
- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1 (20 points)

Calvin and Hobbes play independent games. Calvin wins each game with probability p, and Hobbes wins with probability $q = 1 - p$, where $0 < p < 1$. The match ends as soon as either player has won two more games than the other.

(a) Let T be the total number of games played. Find $\Pr(T = 2k)$, $k = 1, 2, \ldots$. *Hint:* Consider the games in consecutive pairs.

(b) Find $\mathrm{E}[T]$. Which value of p maximizes this expectation?

(c) Find the probability that Calvin wins the match.

Solution

(a) At the start of a pair, the scores are tied. A split pair (CH or HC) has probability $2pq$ and restores the tie without ending the match; a pair CC or HH ends the match, with total probability $p^2 + q^2$. Thus the match can only end after an even number of games, and independence gives

$$\boxed{\Pr(T = 2k) = (2pq)^{k-1}(p^2 + q^2), \qquad k = 1, 2, \ldots.}$$

(b) The number of pairs $T/2$ is geometric on $\{1, 2, \ldots\}$ with success probability $p^2 + q^2$. Hence

$$\boxed{\mathrm{E}[T] = \frac{2}{p^2 + q^2} = \frac{2}{2(p - \frac{1}{2})^2 + \frac{1}{2}}.}$$

The denominator is minimized uniquely at $\boxed{p = 1/2}$, where the maximum expected duration is $\boxed{4}$ games.

(c) Calvin wins exactly when some number of split pairs is followed by CC. Summing these disjoint possibilities gives

$$\boxed{\Pr(\text{Calvin wins}) = \sum_{k=1}^{\infty} (2pq)^{k-1} p^2 = \frac{p^2}{1 - 2pq} = \frac{p^2}{p^2 + q^2}.}$$

Here $2pq \leq 1/2 < 1$, so the geometric series converges and the match ends with probability one.

Problem 2 (20 points)

A box contains 100 shoelaces, each with two loose ends. Repeatedly pick two uniformly random loose ends and tie them together. A tie either joins two open pieces or closes one piece into a loop. Continue until all ends have been tied. Let L be the total number of loops formed.

(a) When m open pieces remain, find the probability that the next tie creates a loop.

(b) Find $\mathrm{E}[L]$ as a finite sum. Independence of the loop indicators is *not* assumed.

(c) Find the probability that only one loop is formed by the end. A finite product is acceptable.

Solution

(a) With m open pieces, there are $2m$ loose ends. Of the $\binom{2m}{2}$ equally likely pairs, exactly m consist of the two ends of the same piece. Therefore, conditional on any previous history,

$$\boxed{\Pr(\text{next tie creates a loop} \mid m \text{ open pieces}) = \frac{m}{\binom{2m}{2}} = \frac{1}{2m-1}.}$$

(b) Each tie reduces the number of open pieces by one, whether it joins two pieces or closes one into a loop. Thus the process visits $m = 100, 99, \ldots, 1$ exactly once. Let J_m indicate that a loop is created at the step with m open pieces. By linearity of expectation,

$$\boxed{\mathrm{E}[L] = \mathrm{E}\left[\sum_{m=1}^{100} J_m\right] = \sum_{m=1}^{100} \frac{1}{2m-1}.}$$

No independence of the indicators is needed.

(c) At $m = 1$, a loop is certain to form. Consequently, $L = 1$ if and only if no loop is created at any of the preceding steps. Applying the chain rule and the history-independent conditional probability in (a),

$$\boxed{\Pr(L = 1) = \prod_{m=2}^{100} \left(1 - \frac{1}{2m-1}\right) = \prod_{m=2}^{100} \frac{2m-2}{2m-1} = \frac{198!!}{199!!}.}$$

Problem 3 (20 points)

A monitoring device is *inactive* on a given day with probability $\alpha \in (0,1)$. On an active day, each of n independent sensors produces an alert with a small probability u. Approximate the active-day alert count by Pois(λ), where $\lambda = nu$ (with large n and small u). Let X be the number of alerts on a randomly selected day. The inactive/active status is independent of the sensor outcomes.

(a) Identify the original distribution of the active-day alert count before approximation. Under the Poisson approximation, find $\Pr(X = 0)$ and $\Pr(X = k)$ for all integers $k \geq 1$.

(b) Find $\mathrm{E}[X]$ and $\mathrm{Var}(X)$. You may introduce independent variables $I \sim \mathrm{Bern}(\alpha)$ and $Y \sim \mathrm{Pois}(\lambda)$ with $X = (1 - I)Y$.

(c) Take $n = 200$, $u = 0.01$ and $\alpha = 1/5$ (so $\lambda = 2$). Compare $\Pr(X = 0)$ under the mixture model with the zero probability under Pois($\mathrm{E}[X]$). Explain briefly why using only the mean to select a Poisson model can be misleading.

Solution

(a) Before approximation, the active-day alert count has distribution $\mathrm{Bin}(n, u)$. Under the Poisson approximation, conditioning on the device status gives the zero-inflated Poisson PMF

$$\boxed{\Pr(X = 0) = \alpha + (1 - \alpha)e^{-\lambda}, \qquad \Pr(X = k) = (1 - \alpha)e^{-\lambda}\frac{\lambda^k}{k!}, \quad k \geq 1.}$$

(b) Let $I \sim \mathrm{Bern}(\alpha)$ and $Y \sim \mathrm{Pois}(\lambda)$ be independent, with $X = (1 - I)Y$. Since $\mathrm{E}[Y] = \lambda$ and $\mathrm{E}[Y^2] = \lambda + \lambda^2$,

$$\mathrm{E}[X] = \mathrm{E}[1 - I]\mathrm{E}[Y] = (1 - \alpha)\lambda,$$
$$\mathrm{E}[X^2] = \mathrm{E}[(1 - I)^2]\mathrm{E}[Y^2] = (1 - \alpha)(\lambda + \lambda^2).$$

Subtracting the square of the mean yields

$$\boxed{\mathrm{E}[X] = (1 - \alpha)\lambda, \qquad \mathrm{Var}(X) = (1 - \alpha)\lambda + \alpha(1 - \alpha)\lambda^2.}$$

These moments refer to the Poisson approximation in (a).

(c) Here $\mathrm{E}[X] = (4/5) \cdot 2 = 8/5$. The two zero probabilities are

$$\boxed{\Pr(X = 0) = \frac{1}{5} + \frac{4}{5}e^{-2} \approx 0.30827, \qquad \Pr(\mathrm{Pois}(8/5) = 0) = e^{-8/5} \approx 0.20190.}$$

Inactivity creates additional zeros. More generally, strict convexity of e^{-x} gives $\alpha + (1-\alpha)e^{-\lambda} > e^{-(1-\alpha)\lambda}$ for $\lambda > 0$. The mixture also has variance 2.24, larger than its mean 1.6, whereas a Poisson variable has equal mean and variance. Thus matching only the mean misses both the excess zeros and the extra variability.

Problem 4 (20 points)

A fair coin is tossed independently $n \geq 5$ times. Let X count the occurrences of the pattern HTH in consecutive positions, allowing overlaps (e.g., HTHTH contains two occurrences). For $i = 1, \ldots, n-2$, let I_i indicate the occurrence starting at position i, so $X = \sum_{i=1}^{n-2} I_i$.

(a) Find $\mathrm{E}[X]$.

(b) Compute $\Pr(I_i = I_{i+1} = 1)$ and $\Pr(I_i = I_{i+2} = 1)$. Explain why I_i and I_j are independent whenever $|i - j| \geq 3$.

(c) Derive $\mathrm{Var}(X)$ as a function of n. Evaluate both $\mathrm{E}[X]$ and $\mathrm{Var}(X)$ when $n = 20$.

Solution

(a) Each length-three window equals HTH with probability $1/8$. Linearity of expectation therefore gives

$$\boxed{\mathrm{E}[X] = \sum_{i=1}^{n-2} \mathrm{E}[I_i] = \frac{n-2}{8}.}$$

(b) Occurrences starting one position apart are incompatible: the shared toss at position $i+1$ would have to be both T and H. Occurrences starting two positions apart require the five-toss string HTHTH. Thus

$$\boxed{\Pr(I_i = I_{i+1} = 1) = 0, \qquad \Pr(I_i = I_{i+2} = 1) = \frac{1}{2^5} = \frac{1}{32}.}$$

For $|i-j| \geq 3$, the two indicators depend on disjoint sets of independent tosses, so they are independent.

(c) Each indicator has variance $(1/8)(7/8) = 7/64$. Using (b),

$$\mathrm{Cov}(I_i, I_{i+1}) = -\frac{1}{64}, \qquad \mathrm{Cov}(I_i, I_{i+2}) = \frac{1}{32} - \frac{1}{64} = \frac{1}{64},$$

and all covariances at larger separations vanish. Among the $n-2$ starting positions, there are $n-3$ unordered pairs at distance one and $n-4$ at distance two. Hence

$$\begin{aligned}\mathrm{Var}(X) &= \sum_{i=1}^{n-2} \mathrm{Var}(I_i) + 2 \sum_{1 \leq i < j \leq n-2} \mathrm{Cov}(I_i, I_j) \\ &= \frac{7(n-2) - 2(n-3) + 2(n-4)}{64} = \boxed{\frac{7n-16}{64}}.\end{aligned}$$

For $n = 20$, this gives

$$\boxed{\mathrm{E}[X] = \frac{9}{4} = 2.25, \qquad \mathrm{Var}(X) = \frac{31}{16} = 1.9375.}$$

Problem 5 (20 points)

A transmitter repeats independent attempts, each successful with probability $p \in (0,1)$. It stops at the *third* successful attempt. Let N be its total number of attempts, and $Y = N - 3$ the number of failures before the third success.

(a) Find the PMF of Y and identify its distribution.

(b) Express $\Pr(N \le 6)$ in terms of a binomial random variable counting successes in the first six attempts. Evaluate this probability when $p = 1/2$.

(c) View Y as a sum of three independent geometric random variables, each counting failures before a success. Find its PGF $G_Y(t) = \mathrm{E}[t^Y]$ and use $G'_Y(1)$ and $G''_Y(1)$ to obtain $\mathrm{E}[N]$ and $\mathrm{Var}(N)$.

Solution

(a) Write $q = 1-p$. The event $Y = y$ requires exactly two successes and y failures in the first $y+2$ attempts, followed by a success. Therefore

$$\boxed{\Pr(Y = y) = \binom{y+2}{2} p^3 q^y, \qquad y = 0, 1, 2, \ldots.}$$

Thus Y is negative binomial with parameters $r = 3$ and p, under the convention that it counts failures before the third success.

(b) Extend the independent attempt sequence beyond the stopping time, and let B count successes in its first six attempts. Then $B \sim \mathrm{Bin}(6, p)$, and $N \le 6$ if and only if $B \ge 3$. Hence

$$\boxed{\Pr(N \le 6) = \Pr(B \ge 3) = \sum_{j=3}^{6} \binom{6}{j} p^j q^{6-j}.}$$

When $p = 1/2$, the probability is

$$\boxed{\frac{20 + 15 + 6 + 1}{64} = \frac{21}{32}.}$$

(c) Let Z_1, Z_2, Z_3 count failures before the first success, between the first and second successes, and between the second and third successes, respectively. Independence of the attempts gives independent Z_j with $\Pr(Z_j = k) = pq^k$ for $k \ge 0$, and $Y = Z_1 + Z_2 + Z_3$. Consequently,

$$G_{Z_j}(t) = \sum_{k=0}^{\infty} pq^k t^k = \frac{p}{1 - qt}, \qquad \boxed{G_Y(t) = \left(\frac{p}{1 - qt}\right)^3}, \quad |qt| < 1.$$

Differentiating and evaluating at $t = 1$ gives

$$G'_Y(1) = \frac{3q}{p}, \qquad G''_Y(1) = \frac{12q^2}{p^2}.$$

Since $G'_Y(1) = \mathrm{E}[Y]$ and $G''_Y(1) = \mathrm{E}[Y(Y-1)]$,

$$\mathrm{E}[N] = 3 + G'_Y(1) = \boxed{\frac{3}{p}},$$

$$\mathrm{Var}(N) = \mathrm{Var}(Y) = G''_Y(1) + G'_Y(1) - [G'_Y(1)]^2$$

$$= \frac{12q^2}{p^2} + \frac{3q}{p} - \frac{9q^2}{p^2} = \boxed{\frac{3(1-p)}{p^2}}.$$